GENERATIVE AI • AI-02 • IDEA SUBMISSION

AirSense AI

Predictive Air Quality Intelligence for Indian Cities

An AI-driven platform that fuses live sensor data, weather signals and city-level pollution trends into forecasts, heatmaps and citizen-ready alerts — moving pollution response from reactive to predictive.

Live Demo: airsense-ai-rho.vercel.app

Team Name: Hydra | *Team Leader: Anubhav Singh*

Built by Anubhav Singh

BACKGROUND

Pollution Monitoring Is Reactive, Not Predictive

No Early Warning

Cities track pollution after it spikes — residents and authorities react only once air quality has already turned hazardous.

Fragmented Signals

Sensor readings, satellite imagery and weather data live in silos, so no single view forecasts what's coming next.

Citizens Left Uninformed

Vulnerable groups (children, elderly, asthmatics) rarely get timely, personalised alerts before exposure happens.

Limited Policy Tools

Government bodies lack ready analytics to identify hotspots early and target mitigation where it matters most.

Problem Statement: Develop an AI-driven platform combining satellite imagery, weather information and sensor data to forecast air quality, identify pollution hotspots, and recommend mitigation strategies.

PROPOSED SOLUTION

AirSense AI — One Platform, Full Pollution Lifecycle

AirSense AI is a full-stack air quality intelligence platform that turns live monitoring data into forward-looking, actionable intelligence for citizens, businesses and government bodies alike — for 50+ Indian cities today, and expanding.

50+ Indian cities covered with live AQI data

Ingest

Live AQI + pollutant data pulled from the WAQI network across cities

Analyse

AI models & Gemini-powered reasoning read trends and context

Forecast

7-day AQI outlook, pollution hotspots & risk bands

Act

Citizen alerts, PDF reports & government-ready analytics

KEY FEATURES

What AirSense AI Delivers Today

Live AQI Dashboard

Real-time air quality pulled from the WAQI (World Air Quality Index) API across 50+ Indian cities.

7-Day AQI Forecasting

Predictive readouts modelled on historical and meteorological trends.

Pollution Heatmaps

Zone-level PM2.5 / PM10 visualisation, refreshed hourly.

Smart Citizen Alerts

Threshold-based notifications the moment air quality crosses a risk band.

AirBot — AI Chatbot

Google Gemini-powered assistant answering air quality & health queries in natural language.

Downloadable PDF Reports

On-demand pollutant breakdowns with health guidance, generated in-app.

Secure Authentication

Google & GitHub OAuth login via NextAuth.js.

Premium Payments

Razorpay-integrated checkout for premium reports & subscriptions.

Automated Email Delivery

Reports and alerts delivered straight to inbox via Gmail SMTP.

PRODUCT WALKTHROUGH

Live, Working Demo — Not Just a Concept

AirSense AI is already deployed and functional. The MVP is live on Vercel with real API integrations — not mockups.

https://airsense-ai-rho.vercel.app

Dashboard

City search, live AQI gauge & pollutant breakdown

Forecast & Trends

7-day AQI outlook charted with Recharts

Alerts Centre

Personalised, threshold-based risk notifications

AirBot Chat

Gemini-powered conversational air-quality assistant

Reports

One-click PDF pollutant report generation & email delivery

Account & Billing

OAuth login, profile, and Razorpay premium upgrade

Technology Stack — What's Actually Running

Frontend Framework

Next.js 16 (App Router) · React 19

Styling / UI

Tailwind CSS v4 · Lucide & Simple-Icons icon sets

Backend

Next.js API Routes (Node.js serverless functions)

Database

MongoDB with Mongoose ODM — User, City, Report & Alert models

Authentication

NextAuth.js — Google & GitHub OAuth

AI / GenAI

Google Gemini API (@google/genai) — powers AirBot assistant

External Data API

WAQI (World Air Quality Index) live sensor network

Payments

Razorpay — order creation & signature verification

PDF Generation

@react-pdf/renderer — dynamic pollutant reports

Email Delivery

Nodemailer via Gmail SMTP

Hosting / Deployment

Vercel (CI/CD from GitHub)

Extra Backend Service

Firebase & Firebase Admin SDK

System Architecture

Layer	Components
Client Layer	• Next.js App Router pages — Dashboard, Forecast, Alerts, Reports, Account, Login • Reusable UI components — Gauge, SpectrumRibbon, ChatWidget, Header/Footer
Application / API Layer	• Serverless API routes — /api/city/[name], /api/chat, /api/alerts, /api/reports • /api/payment/create-order & verify, /api/auth (NextAuth), /api/user
Data & Integration Layer	• MongoDB (Mongoose models) for users, cities, reports & alerts • WAQI API · Google Gemini API · Razorpay · Firebase · Gmail SMTP

Deployed on Vercel with CI/CD from GitHub · Environment-based secrets for all third-party API keys

INNOVATION & UNIQUENESS

What Sets AirSense AI Apart

Conversational AI Layer

AirBot uses Google Gemini to translate raw AQI numbers into plain-language health guidance — not just a chart.

Forecast-First, Not Reactive

7-day predictive outlook shifts the platform from after-the-fact reporting to preventive planning.

Actionable Reports Built In

Auto-generated PDF reports with health recommendations, delivered by email — ready for sharing with authorities or family.

Dual Audience by Design

Same data pipeline serves a simple citizen alert experience and a government-grade analytics view.

TARGET USERS

Who Benefits From AirSense AI

Everyday Citizens

Real-time AQI, personalised alerts and health tips for daily planning — especially for children, elderly & asthmatic residents.

Government & Municipal Bodies

Hotspot analytics and forecast data to plan targeted mitigation and public advisories.

Urban Planners & NGOs

Zone-level heatmaps to prioritise interventions and track long-term air quality trends.

Businesses & Institutions

Downloadable compliance-ready reports for schools, offices and construction sites.

Built to Grow Beyond the MVP

50+ Indian cities live on the dashboard today

7-Day Forward-looking AQI forecast window

24/7 Automated monitoring & alerting

Scalability Path

- Serverless Next.js + MongoDB architecture scales horizontally with city and user growth
- Modular data layer makes it straightforward to add satellite imagery & IoT sensor feeds
- Multi-tenant ready — same platform can license to municipal corporations & enterprises
- Vercel deployment enables instant global edge delivery as adoption expands beyond India

In Active Development — Roadmap Ahead

AirSense AI is a live, working MVP under active development. Core modules — dashboard, forecasting, alerts, AirBot and reports — are functional today, with further refinement and features shipping in upcoming iterations.

Satellite Imagery Integration

Ingest satellite-derived pollution data to strengthen the forecasting model per the problem statement.

Deeper ML Forecasting

Move from trend-based prediction to a trained time-series ML model for higher forecast accuracy.

Mobile App

Native mobile experience for push-based citizen alerts on the go.

Government Analytics Console

Dedicated dashboard for municipal bodies with exportable hotspot analytics.

Hyperlocal Heatmaps

Higher-resolution, street-level pollution mapping using densified sensor + satellite input.

Multi-Country Expansion

Extend the WAQI-based pipeline beyond India to other high-pollution regions.

Thank You

From reactive monitoring to predictive, preventive air quality intelligence.

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Team Leader: Anubhav Singh

Team Members: Aradhya Pandey, Aaradhy Sinha